

02456 - DEEP LEARNING PROJECT - GROUP 8

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ABSTRACT

Experimental mapping of cellular responses to genetic perturbations cannot cover the full combinatorial space of gene interactions. We train a conditional Flow Matching model on Virtual Cell Challenge data, operating in 4,721-dim $\log_1 p$ gene-expression space with ESM-2 embeddings of the target gene as conditioning, enabling zero-shot prediction for held-out perturbations. The model clears the cell-mean baseline on perturbation discrimination (PDS-VCC +11.5%) and differential-expression recovery (DES +20.6%) on a held-out test set, but underperforms on Pearson r and pseudobulk MAE – it identifies *which* genes respond more accurately than *how*.

1. INTRODUCTION

Understanding cellular responses to genetic perturbations is fundamental to biology, yet experimentally mapping the vast combinatorial space of gene interactions using techniques like CRISPR interference and single-cell RNA sequencing (scRNA-seq) is prohibitively expensive. The Virtual Cell Challenge 2025 [1] aims to address this. The challenge gives us $\sim 183,000$ single-cell profiles of human embryonic stem cells (H1 hESC) with 150 CRISPRi knockdowns, and asks for models that generalise to perturbations they have never seen during training.

We approach this with Flow Matching [2]: we learn a deterministic velocity field that transports a control cell to its perturbed state directly in $\log_1 p$ gene-expression space. Working in expression space, rather than in a latent embedding such as scVI [3] or STATE’s SE-600M [4] (both of which we explored), keeps the loss biologically interpretable and aligned with the evaluation metrics. Each perturbation is identified only by its target gene, which we represent with a 480-dim ESM-2-35M [5] protein embedding, which we found to outperform the higher-dimensional GENEPT-3072 [6] embeddings. The velocity network, FLOWNET, is a residual MLP with Adaptive Layer Normalization (AdaLN) conditioning.

Three design choices proved decisive: Hungarian optimal-transport pairing within mini-batches to straighten flows, a per-perturbation mean-LFC (Log Fold Change) auxiliary loss to complement the per-cell velocity objective, and gene-activity weighting to concentrate gradient on differentially

expressed genes.

2. MATERIALS AND METHODS

Github: <https://github.com/DhruvaSakhare/geneFlow-deepLearning>

2.1. Dataset and preprocessing

The Virtual Cell Challenge dataset [1] contains $\sim 183,000$ H1 hESC cells across 150 CRISPRi perturbations. Raw counts are normalised to 10,000 transcripts per cell and $\log_1 p$ -transformed. From the resulting expression matrix we retain the union of the top-200 most differentially expressed genes per training perturbation (4,721 genes total). There are a total of 38,000 control cells. The control reference $\bar{x}_{\text{ctrl}}$ is the mean profile of non-targeting cells.

2.2. Perturbation conditioning

Each perturbation is identified only by its target-gene name. We embed each target with ESM-2-35M [5]: protein sequences are retrieved from UniProt and mean-pooled across residues into a 480-dim vector. In comparison, GENEPT-3072 embeddings [6] performed worse.

2.3. Flow Matching

Flow Matching [2] learns a deterministic velocity field whose ODE transports one distribution into another. Given a control cell x_0 and a paired perturbed cell x_1 , the linear interpolation $x_t = (1 - t)x_0 + tx_1$ with $t \sim \mathcal{U}(0, 1)$ has target velocity $u = x_1 - x_0$, and the network $v_\theta(x_t, t, y)$ is trained to predict it, where $y \in \mathbb{R}^{480}$ is the ESM-2 embedding of the targeted gene. We add three components to the vanilla setup:

Hungarian Optimal Transport (OT) pairing Each mini-batch contains 16 perturbations $\times$ 16 cells (batch 256). Within each perturbation group we solve a 16×16 L_2 assignment problem; the resulting pairing minimises total trajectory length $\sum_i \|x_1^{(i)} - x_0^{(i)}\|$, giving straighter flows that are easier to learn.

Gene-activity-weighted loss Per-perturbation LFC is sparse: each perturbation moves only a small fraction of the

4,721 selected genes. We therefore take $\mathcal{L}_{\text{FM}}$ to be a gene-weighted MSE,

$$\mathcal{L}_{\text{FM}} = \sum_g w_g (v_\theta(x_t, t, y)_g - u_g)^2, \quad w_g = \overline{|u_g|},$$

where $\overline{|u_g|}$ is the mean absolute velocity for gene g across training, focusing gradient on genes that actually change.

Mean-LFC auxiliary loss For each perturbation p in the batch we add the squared distance between predicted and true mean velocity vectors:

$$\mathcal{L} = \mathcal{L}_{\text{FM}} + \lambda_{\text{LFC}} \sum_p \|\overline{v}_{\theta p} - \overline{u}_p\|_2^2, \quad \lambda_{\text{LFC}} = 0.3.$$

Larger λ_{LFC} raised Pearson r but homogenised predictions across perturbations and hurt PDS.

Algorithm 1 Flow Matching training step

- 1: Pair control and perturbed cells via Hungarian OT (per perturbation)
 - 2: Sample $t \sim \mathcal{U}(0, 1)$, set $x_t = (1 - t)x_0 + tx_1$
 - 3: Predict $\hat{v} = v_\theta(x_t, t, y)$; target $u = x_1 - x_0$
 - 4: $\mathcal{L} = \underbrace{\sum_g w_g (\hat{v}_g - u_g)^2}_{\text{gene-weighted FM}} + \lambda_{\text{LFC}} \underbrace{\|\hat{\bar{v}} - \bar{u}\|_2^2}_{\text{mean-LFC aux}}$
 - 5: AdamW step on θ , where θ denotes the trainable parameters of the velocity network.
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2.4. FlowNet architecture

The velocity network v_θ (Figure 1) compresses the cell x_t into a 512-dim hidden state h , refines it through 8 residual blocks, and expands back to 4,721 dims to predict the velocity. Conditioning on time t and the ESM-2 gene embedding is handled via a 1024-dim vector c that modulates every block through AdaLN,

$$\text{AdaLN}(h, c) = \text{LayerNorm}(h) (1 + \gamma_c) + \beta_c,$$

so each block updates the stream as $h \leftarrow \text{MLP}(\text{AdaLN}(h, c))$. The output head is zero-initialised so v_θ emits zero velocity at the start of training, making the initial flow an identity map. Total parameters: $\sim 32.8\text{M}$.

2.5. Inference

Given a control cell x_0 and gene embedding y , we integrate v_θ over $t \in [0, 1]$ with a fixed-step fourth-order Runge–Kutta scheme (50 steps); the integrated endpoint x_1 is the predicted perturbed cell directly in $\log_1 p$ space. For each held-out perturbation we sample 1,000 control cells and aggregate predictions for population-level metrics.

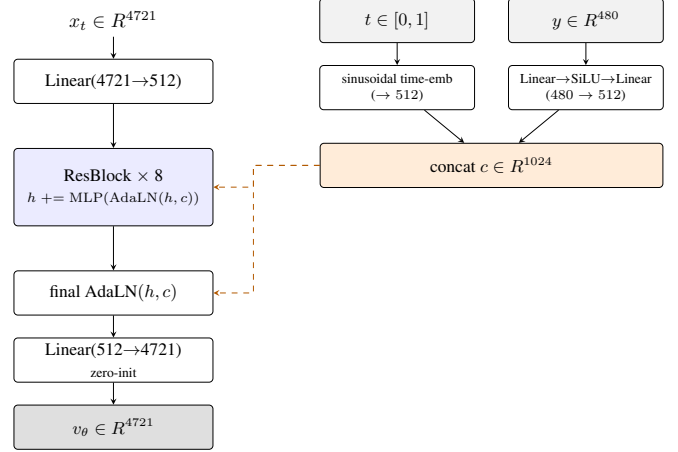


Fig. 1. FlowNet architecture. The gene-expression state x_t flows through 8 AdaLN-conditioned residual blocks (left column). Time and the ESM-2 gene embedding are projected to 512-dim each and concatenated into a 1024-dim conditioning vector c (right column) that modulates every block and the final AdaLN via dashed arrows. The output head is zero-initialised so v_θ emits zero velocity at the start of training, making the initial flow an identity map.

2.6. Splits and optimisation

2.6.1. Splits

The 150 perturbations are split by gene into 120 train, 15 validation, 15 test, so every held-out perturbation is unseen during training.

2.6.2. Optimisation.

100 epochs of AdamW [7] with $\text{lr} = 10^{-4}$ and a cosine schedule.

2.7. Evaluation metrics

We use four metrics, all computed on log-fold-change vectors $\delta = \overline{x_{\text{pert}}} - \overline{x_{\text{ctrl}}}$ in $\log_1 p$ space.

Pearson r between predicted and true LFC vectors over all 4,721 panel genes – picks up both sign and magnitude.

PDS-VCC For each predicted LFC we rank all true LFCs by L_1 distance (target gene column excluded) and record where the correct match lands. PDS-VCC averages $1 - (\text{rank} - 1)/N$ across the N perturbations – 1 if every prediction picks its own true partner first, and $(N + 1)/(2N) = 0.533$ for $N = 15$ when all predictions are identical.

DES Per-gene Mann–Whitney U test of perturbed vs control, run separately on predicted and true cells, with Benjamini–Hochberg at FDR 0.05 to get DE gene sets

G_{pred} and G_{true} . If $|G_{\text{pred}}| > |G_{\text{true}}|$ we keep only the top- $|G_{\text{true}}|$ predicted genes by $|\delta_{\text{pred}}|$; DES is then $|G_{\text{pred}} \cap G_{\text{true}}|/|G_{\text{true}}|$. It rewards getting the right genes, ignoring direction and magnitude.

MAE Mean absolute error between predicted and true per-gene pseudobulk means in $\log_1 p$ space, $\frac{1}{G} \sum_g |\bar{x}_{\text{pred}_g} - \bar{x}_{\text{true}_g}|$.

VCC composite The leaderboard score is the average of PDS-VCC, DES and MAE, each first scaled against the cell-mean baseline. Full details: <https://virtualcellchallenge.org/evaluation>.

2.7.1. Checkpoint selection.

We evaluate on validation every 5 epochs and select the best checkpoint by the VCC-aligned score

$$S = r_{\text{Pearson}} + (\text{PDS}_{\text{VCC}} - 0.533),$$

where $0.533 = (N + 1)/(2N)$ is the structural PDS-VCC for $N = 15$ identical predictions.

3. RESULTS

3.1. Cell-mean baseline

The cell-mean baseline predicts the same training-LFC vector $\bar{\delta} = \frac{1}{|P_{\text{train}}|} \sum_{p \in P_{\text{train}}} (\bar{x}_p - \bar{x}_{\text{ctrl}})$ for every held-out perturbation. Because CRISPRi knockdowns share a stress-response signature, this is a non-trivial reference (Table 1).

Table 1. Cell-mean baseline on the 15 validation perturbations.

Metric	Baseline
Pearson r	0.238
PDS-VCC	0.533
DES	0.138
MAE	0.034

The 0.533 PDS-VCC baseline is structural: with N identical predictions the formula $1 - (\text{rank} - 1)/N$ averages to $(N + 1)/(2N) = 0.533$ for $N = 15$, approaching 0.5 as N grows.

3.2. Validation performance

Training was stable for the full run; the FM loss decreased monotonically while validation metrics peaked around epoch 45 and then oscillated (Figure 2). This decoupling justifies selecting checkpoints by S rather than by validation loss. Table 2 reports best-checkpoint metrics; Δ -baseline uses the simple relative change $(\text{model} - \text{baseline})/\text{baseline}$ for r , and the VCC-style scaling $(\text{model} - \text{baseline})/(1 - \text{baseline})$ for PDS-VCC and DES.

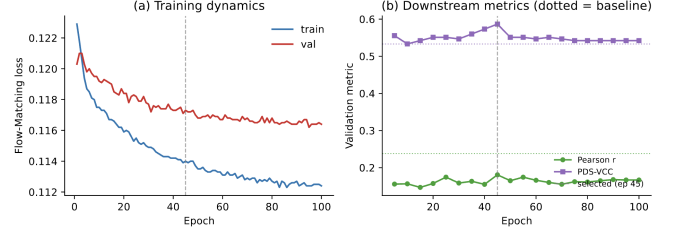


Fig. 2. Training dynamics. **(a)** Train and validation FM loss across the run. **(b)** Validation perturbation-level metrics (Pearson r , PDS-VCC); dotted horizontal lines mark the cell-mean baseline. The selected checkpoint (epoch 45) is highlighted. FM loss keeps decreasing while perturbation metrics peak and then oscillate.

Table 2. Validation metrics at the selected checkpoint (epoch 45).

Metric	Baseline	Model	Δ
Pearson r	0.238	0.181	−24%
PDS-VCC	0.533	0.587	+11.6%
DES	0.138	—	—
MAE	0.034	0.045	clip 0

The model clears the baseline on discrimination (PDS-VCC) by a substantial margin, but underperforms on per-gene magnitude (r) and pseudobulk MAE. DES is evaluated separately on the held-out test set (Table 3). The model identifies *which* perturbations are distinct from one another more reliably than the baseline, yet does not always recover the correct direction and magnitude of individual gene changes.

3.3. Held-out test set

We evaluate the selected checkpoint on the 15 test perturbations – CASP2, CHMP3, CREG1, FDPS, HMGB2, INSIG1, MBTPS1, RAF1, SALL4, SMAGP, SNCA, TSC22D4, TWF2, UQCRQ, USP22 – none of which were used for training or checkpoint selection. The cell-mean baseline is recomputed on the test split (Table 3); both r and DES baselines shift slightly relative to validation because they are scored against a different perturbation set, while PDS-VCC stays at its structural 0.533.

Table 3. Held-out test performance at the selected checkpoint (epoch 45). Baseline recomputed on the test split.

Metric	Baseline	Model	Δ
Pearson r	0.237	0.167	−30%
PDS-VCC	0.533	0.587	+11.5%
DES	0.157	0.331	+20.6%
MAE	0.027	0.041	clip 0

Test performance closely mirrors validation: PDS-VCC is identical (0.587 on both splits). Pearson r remains below

baseline, and MAE clips to zero. The absence of a generalisation gap confirms the selected checkpoint is not overfitted to validation perturbations. The VCC composite – scaling DES and PDS-VCC against baseline and clipping MAE at zero – yields $(20.6 + 11.5 + 0)/3 \approx +10.7\%$ over baseline on the held-out test set.

3.4. Per-perturbation analysis

Performance varies sharply across the 15 test perturbations (Table 4). Two perturbations (MBTPS1, UQCRQ) reach $r > 0.5$ with $DES > 0.55$, and HMGB2 follows at $r = 0.39$, $DES = 0.46$. At the other end, four targets (USP22, CHMP3, SMAGP, FDPS) show a *set-correct, direction-wrong* failure: DES between 0.41 and 0.59 while r is near zero or negative. USP22 is the most extreme ($DES = 0.59$, $r = -0.15$). The model recovers most of the responding gene set but in the wrong direction – it is pretty good at predicting *which* genes are affected, but struggles with *how* (the direction). Performance is better on perturbations whose targets have close relatives in the training set: the ESM-2 embedding lets the model transfer learned effects from a similar gene to the new one.

Table 4. Per-perturbation results on the held-out test set (epoch 45).

Pert	r	DES	Pert	r	DES
MBTPS1	+0.650	0.673	TSC22D4	+0.109	0.225
UQCRQ	+0.578	0.561	SNCA	+0.095	0.207
HMGB2	+0.391	0.461	TWF2	+0.043	0.000
SALL4	+0.252	0.145	SMAGP	+0.029	0.414
INSIG1	+0.178	0.209	CHMP3	-0.027	0.555
RAF1	+0.173	0.236	CREG1	-0.110	0.046
CASP2	+0.152	0.093	USP22	-0.145	0.587
FDPS	+0.136	0.561			
Mean: $r = +0.167$, $DES = 0.331$, $PDS-VCC = 0.587$					

4. DISCUSSION

4.1. What the model learns

The split across metrics is structural, not noise: PDS-VCC and DES reward identifying *which* perturbations and *which* genes are engaged, and on both the model clears baseline; Pearson r and MAE reward direction and magnitude, where it does not. Meaning that the model is better at predicting *which* genes change but not *how*. Some perturbations give better prediction results as the genes might have similar mechanisms with the training set that was captured by the ESM-2 embeddings. The predictions can be improved by expanding the training set to include more perturbations and use multiple embeddings, not just ESM-2 to learn the gene effects not captured by ESM-2.

4.2. Why MAE is below baseline

The baseline wins MAE by being conservative. Its per-gene LFC is the mean of 120 training perturbations, so on genes that are silent for any one perturbation its prediction is essentially zero. The flow model is per-cell: it pushes individual cells through a velocity field, which introduces per-cell noise that does not fully cancel in the pseudobulk mean. On silent genes the model adds wiggle, the baseline adds almost nothing – the baseline wins MAE almost automatically. This is why MAE is the one metric where the baseline is genuinely hard to beat.

4.3. Design choices that mattered

- **Hungarian OT pairing within batches** shortened trajectories and yielded a smoother velocity field.
- **Gene-activity weighting** prevented the near-silent gene majority from dominating gradients, since most of the 4,721 panel genes are unaffected by any single perturbation.
- $\lambda_{LFC} = 0.3$ was the sweet spot: $\lambda_{LFC} = 1.0$ lifted Pearson r but homogenised predictions across perturbations and hurt PDS.

4.4. Methodological caveats

With only 15 validation perturbations, per-perturbation metrics are noisy and the PDS-VCC baseline is structurally inflated to 0.533 (vs. 0.514 on the larger leaderboard test set). Hyperparameters were tuned on validation, so val numbers may be slightly optimistic; the held-out test numbers (Table 3) are the unbiased estimate.

4.5. Future work

- **Pretraining on Replogle 2022** [8]. The genome-scale CRISPRi Perturb-seq atlas ($\sim 2,000+$ essential-gene perturbations in K562) would expose the model to indirect and metabolic effects that pure sequence conditioning cannot recover from a 120-perturbation training set.
- **Cross-attention conditioning.** AdaLN applies a single global scale/shift per block, which may be too coarse for fine-grained perturbation specificity. Cross-attention would let each gene attend to the conditioning vector independently.
- **Multiple gene embeddings.** ESM-2 only captures sequence-level signal, missing indirect/metabolic effects. Combining ESM-2 with other embeddings (e.g. GENEPT) could recover what sequence alone cannot.

DECLARATION OF USE OF GENERATIVE AI

- We have used generative AI tools: yes

List the generative AI tools you have used:

- ChatGPT 5.1
- GitHub Copilot (model Gemini 2.5 Pro)

What did you use the tool(s) for? The tools were used for code debugging, exploring alternative implementation approaches, and performing high-level reasoning checks during model development.

At what stage(s) of the process did you use the tool(s)?

They were used throughout development, including during coding, model setup, and refinement.

How did you use or incorporate the generated output?

All AI-generated suggestions were manually reviewed, verified for correctness, and adapted before use. Core research ideas, analysis, and conclusions were produced independently by the authors.

References

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